# FAST

## National University of Computer & Emerging Sciences

**FAST-NUCES, Lahore Campus**

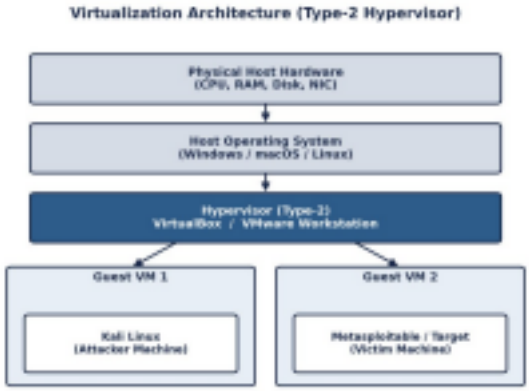
Department of Cyber Security

### Course: Cyber Security

BS Cyber Security — 3rd Semester

# LAB MANUAL

## Lab No. 01 — Virtualization Fundamentals: Installing Kali Linux and Exploring the Linux Command Line




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# 1. Lab Objectives

This is the very first lab of the Cyber Security course, and it lays the practical groundwork for every lab that follows for the rest of the semester. Cyber Security is fundamentally a hands-on discipline — reading about an attack or a defense mechanism is no substitute for actually configuring, breaking, and fixing a real system. Because most offensive and defensive tools used

in the industry run on Linux, and because experimenting with security tools directly on a personal or institute-owned computer is risky and can violate policy, professionals rely on virtualization to build safe, disposable, and fully isolated lab environments. This lab teaches exactly that skill.

By the end of this lab, students will be able to:

- Explain what virtualization is and why it is the foundation of every cyber security lab environment.
- Differentiate between Type-1 and Type-2 hypervisors, and compare VirtualBox and VMware Workstation.
- Install and configure a Type-2 hypervisor on their host machine.
- Correctly download, verify, and install Kali Linux as a guest virtual machine.
- Configure virtual hardware resources (RAM, CPU, storage, networking) appropriate for security testing.
- Take and restore VM snapshots to safely experiment without risk of permanent damage.
- Navigate the Kali Linux desktop environment and understand its purpose-built toolset.
- Use the Linux command-line shell confidently, including navigation, file operations, permissions, package management, and process/user management.
- Apply these fundamentals to complete hands-on exercises and self-assess their own understanding.

**Why this matters:**

Every remaining lab in this course — packet sniffing, vulnerability scanning, exploitation, web application testing, digital forensics — assumes you already have a working Kali Linux VM and are comfortable at the terminal. Treat this lab as the foundation the rest of the semester is built on, not a formality to rush through.



# 2. Introduction & Theoretical

## Background

### 2.1 What is Cyber Security?

Cyber Security is the practice of protecting systems, networks, applications, and data from unauthorized access, disruption, modification, or destruction. It spans multiple domains, including network security, application security, operating system security, cryptography, digital forensics, incident response, and governance/compliance. A recurring theme across all of these domains is that defenders must think like attackers — a principle often summarized as the offensive mindset. To understand how a firewall can be bypassed, how a web application can be exploited, or how malware behaves, a security professional needs a safe environment where such techniques can be practiced legally and without consequence. That environment is what this lab builds.

## 2.2 Why Kali Linux?

Kali Linux is a Debian-based Linux distribution maintained by Offensive Security, purpose-built for penetration testing, digital forensics, and security auditing. Unlike a general-purpose operating system, Kali ships with hundreds of pre-installed and pre-configured security tools — network scanners, vulnerability assessment frameworks, password-cracking utilities, wireless auditing tools, web application testing proxies, and forensic utilities — organized into logical categories. Throughout this course, Kali Linux will serve as our primary "attacker" workstation, from which we will study and practice the tools and techniques covered in each lab.

## 2.3 Understanding Virtualization

Virtualization allows a single physical computer (the host) to run one or more independent, isolated "virtual" computers (guests) simultaneously. A software layer called a hypervisor is responsible for creating and managing these virtual machines (VMs), allocating a share of the host's CPU, memory, storage, and network resources to each one. For a cyber security lab, virtualization is essential for three main reasons:

- Isolation & Safety: Malware, exploits, and misconfigurations inside a VM cannot directly affect the host operating system or other machines on the network, as long as networking is configured correctly.
- Disposability: A VM that becomes corrupted, infected, or misconfigured during an experiment can be deleted and recreated in minutes, or instantly rolled back using a snapshot.
- Realistic multi-machine scenarios: Multiple VMs (e.g., an attacker machine and a vulnerable target machine) can be run at the same time on one laptop and connected through a private virtual network, simulating a real environment.

Hypervisors are broadly classified into two types. A Type-1 ("bare-metal") hypervisor runs directly on the host's hardware without an underlying operating system (examples: VMware ESXi, Microsoft Hyper-V, Proxmox) and is typically used in enterprise data centers. A Type-2 ("hosted") hypervisor runs as an application on top of a conventional host operating system such as Windows, macOS, or Linux (examples: Oracle VirtualBox, VMware Workstation/Fusion). For a



student lab running on a personal laptop, a Type-2 hypervisor is the appropriate and expected choice.

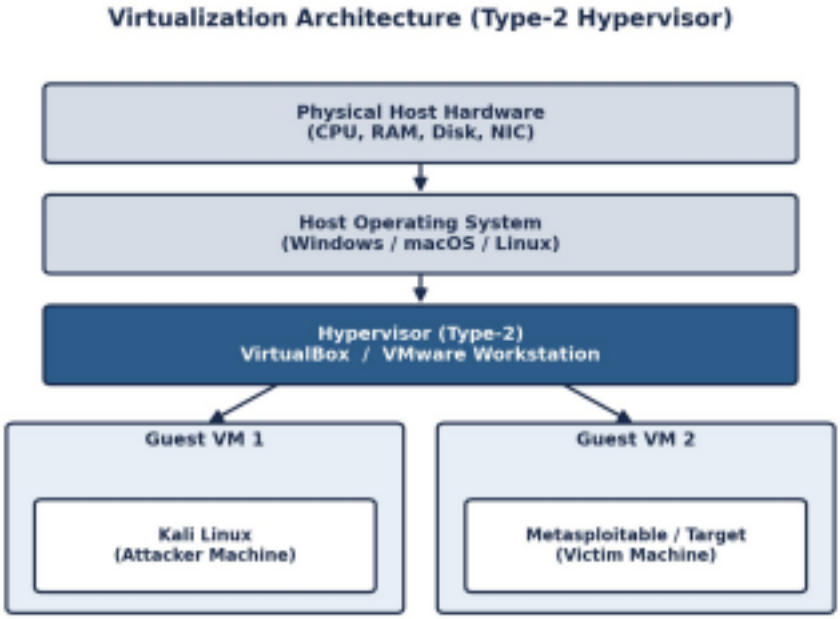



*Figure 2.1 — Type-2 hypervisor architecture: the hypervisor runs as a program inside the host OS and hosts one or more isolated guest VMs.*

## 2.4 VirtualBox vs VMware Workstation

Both tools are Type-2 hypervisors capable of running Kali Linux comfortably. Students may use either tool for this course unless an instructor specifies otherwise; the steps in this manual are described generically and apply to both with only minor interface differences.

| | | |
|---|---|---|
| Cost | Free & Open-Source | Player: free for personal use. Pro: paid |
| Platform Support | Windows, macOS, Linux | Windows, Linux (macOS: VMware Fusion) |
| Performance | Good for general use | Generally smoother 3D/graphics performance |
| Snapshots | Supported | Supported |
| Guest Tools | Guest Additions | VMware Tools |
| Recommended For | Beginners, cross-platform labs | Users needing higher performance/graphics |

# 3. Pre-Lab Requirements

### *Hardware Requirements*

- A 64-bit host CPU with virtualization extensions enabled in BIOS/UEFI (Intel VT-x or AMD V).
- Minimum 8 GB of system RAM (4 GB will be allocated to the VM; 4 GB should remain for the host).
- At least 60 GB of free disk space (SSD strongly recommended for install speed and responsiveness).

### *Software Requirements*

- A hypervisor: Oracle VirtualBox (with the Extension Pack) or VMware Workstation Player/Pro.
- Official Kali Linux installer ISO or pre-built VM image, downloaded only from the official source: https://www.kali.org/get-kali/
- A stable internet connection for downloading software and post-install updates.

### *Knowledge Prerequisites*

- Basic familiarity with operating a computer (file management, installing software).
- No prior Linux experience is required — this lab starts from zero.

**Safety Reminder:**

Only download Kali Linux from the official website (kali.org). Third-party mirrors or "pre-cracked" images circulating online may be tampered with and can contain malware. Always verify the SHA256 checksum of the downloaded ISO against the value published on the official download page before use.





# 4. Lab Activity 1 — Installing the Hypervisor

This activity installs the Type-2 hypervisor that will host your Kali Linux virtual machine. Steps are given for VirtualBox; VMware Workstation users should substitute the equivalent VMware installer and follow the same logical flow.

1. Navigate to https://www.virtualbox.org/wiki/Downloads in your host machine's browser.
2. Download the installer package that matches your host operating system (Windows hosts / macOS hosts / Linux distribution packages).
3. Run the downloaded installer with administrator/root privileges and accept the default installation options, including the network interface driver installation prompt (this is required for VM networking).

4. Once installation completes, launch VirtualBox and confirm the "Oracle VM VirtualBox Manager" window opens without errors.
5. Download and install the matching VirtualBox Extension Pack from the same downloads page — this adds USB 3.0 support, disk encryption, and other features useful for later labs.

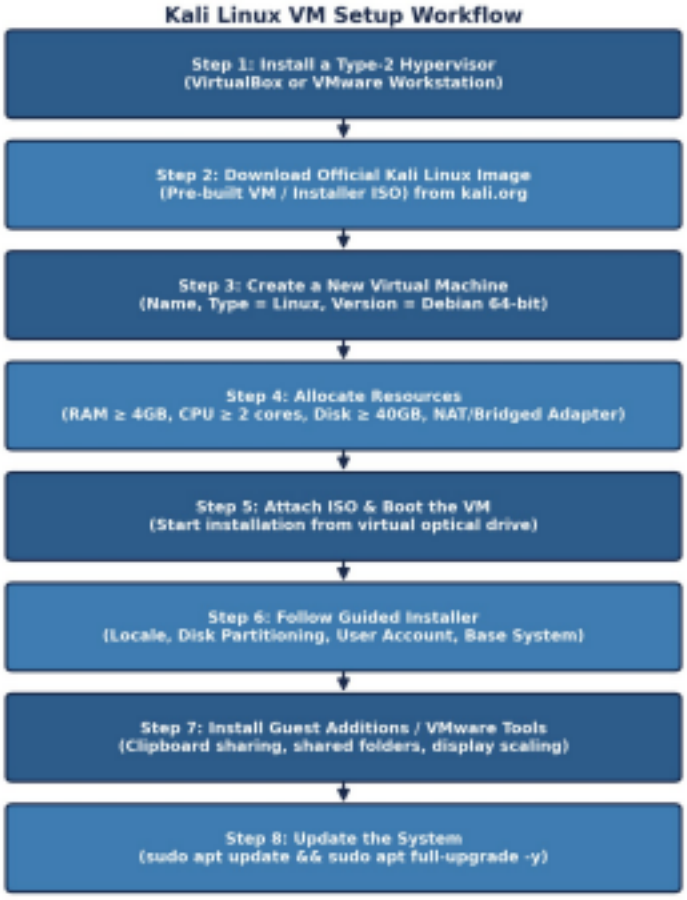



*Figure 4.1 — Overall workflow this and the following activities will walk through, from hypervisor installation to a fully updated Kali VM.*



## 5. Lab Activity 2 — Obtaining Kali Linux

1. Go to the official Kali Linux downloads page: https://www.kali.org/get-kali/
2. Select "Virtual Machines" if you prefer a pre-built VM image (fastest option, ready to import), or "Installer" if you prefer to build the VM manually and install from ISO (recommended for learning, and the path this manual follows).
3. Choose the 64-bit installer ISO (Kali Linux Installer, amd64).
4. After the download completes, verify file integrity by generating the SHA256 checksum of

the downloaded file and comparing it against the value listed on the Kali download page.

```
# On Windows (PowerShell):
Get-FileHash .\kali-linux-2025.x-installer-amd64.iso -Algorithm SHA256

# On macOS / Linux:
sha256sum kali-linux-2025.x-installer-amd64.iso
```

If the checksum does not match, delete the file and re-download it — do not proceed with an unverified image.



## 6. Lab Activity 3 — Creating and Configuring the Virtual Machine

1. Open VirtualBox Manager and click "New".
2. Name the VM (e.g., "Kali-Linux-CySec-Lab"); VirtualBox should auto-detect Type = Linux, Version = Debian (64-bit) from the name — set it manually if it does not.

3. Allocate Memory (RAM): at least 4096 MB. Leave sufficient RAM for the host OS to remain responsive.
4. Create a Virtual Hard Disk: choose "Create a virtual hard disk now", use the VDI (VirtualBox Disk Image) format, select "Dynamically allocated", and set the size to at least 40 GB (60–80 GB is preferable for later labs that install additional tools).
5. Under System, allocate at least 2 CPU cores (Settings System Processor). → →
6. Under Display, increase Video Memory to the maximum allowed and enable 3D Acceleration if available.
7. Under Storage, click the empty optical drive entry and attach the downloaded Kali ISO file.
8. Under Network, confirm Adapter 1 is Enabled and set to NAT (simplest option — the VM can reach the internet but is not directly reachable from the external network). Later labs may ask you to change this to a Host-Only or Internal Network when a private attacker/target lab topology is required.

**Tip — Snapshots:**

Once the VM is installed and updated (end of Activity 5), take a Snapshot (right-click the VM → Snapshot Take) named "Clean Install". Before trying anything risky in future labs (e.g., running → an exploit, editing system files), take a new snapshot. If something breaks, you can restore to a working state in seconds instead of reinstalling.



## 7. Lab Activity 4 — Installing Kali Linux

Start the VM. It will boot from the attached ISO into the Kali installer. Follow the guided steps below.

1. At the boot menu, select "Graphical Install" (recommended for first-time users).
2. Select your language, location, and keyboard layout.
3. Enter a hostname for the machine (e.g., kali) and, optionally, a domain name (leave blank if unsure).
4. Create a full name for the user, a username, and a strong password. Remember this password — you will need it for every login and for any command requiring sudo.
5. Select your time zone.
6. Disk Partitioning: choose "Guided – use entire disk" and then "All files in one partition" (recommended for beginners). Confirm and write the changes to disk when prompted.
7. Software Selection: keep the default Kali metapackages selected (Desktop environment + top10 tools) unless your instructor specifies a minimal install.
8. Install the GRUB bootloader to the primary virtual disk (/dev/sda) when prompted.
9. Finish the installation and reboot when prompted. Remove/eject the ISO from the virtual

optical drive if the VM tries to boot from it again (Devices Optical Drives Remove disk →
→ from virtual drive).

10. Log in with the username and password created during installation.



# 8. Lab Activity 5 — Post-Installation Configuration

### *8.1 Install Guest Additions*

Guest Additions (VirtualBox) or VMware Tools (VMware) enable shared clipboard, drag-and-drop, shared folders, and automatic display resizing between host and guest.

1. With the VM running, go to Devices Insert Guest Additions CD Image. → 2.

Open a terminal inside Kali and run the commands below, then reboot the VM.

```
sudo apt update
sudo apt install -y build-essential dkms linux-headers-$(uname -r) sudo mount
/dev/cdrom /mnt
sudo /mnt/VBoxLinuxAdditions.run
sudo reboot
```

### *8.2 Update the System*

Always update a freshly installed system before use — this pulls the latest security patches and

tool versions.

```
sudo apt update && sudo apt full-upgrade -y
sudo apt autoremove -y
```

### *8.3 Take a Clean Snapshot*

As noted earlier, take a snapshot named "Clean Install" now, before installing or modifying anything further. This becomes your recovery point for the rest of the semester.



## 9. Lab Activity 6 — Exploring the Kali Linux Desktop Environment

Kali's default desktop (Xfce) is organized to give quick access to its security toolset. Spend time exploring the following before moving to the command line, since you will return to these tools throughout the semester.

- Applications Menu (top-left): tools are grouped by category — Information Gathering, Vulnerability Analysis, Web Application Analysis, Password Attacks, Wireless Attacks, Exploitation Tools, Sniffing & Spoofing, Post Exploitation, Forensics, and Reporting Tools.
- Favorites bar: quick-launch icons for commonly used tools such as the Terminal, Firefox, and Files manager.
- Terminal emulator: the single most important application in Kali — nearly every lab in this course will be conducted primarily from the terminal.
- Kali Documentation (accessible from the menu or https://www.kali.org/docs/): the

authoritative reference for tool usage, troubleshooting, and updates.

1. Open the Terminal from the favorites bar or by pressing Ctrl+Alt+T.
2. Run whoami, hostname, and uname -a to confirm your username, machine name, and kernel version.
3. Browse the Applications menu and identify at least three tool categories you expect to use later in this course.



# 10. Lab Activity 7 — Linux Command-Line Fundamentals

Almost every security tool covered in this course is operated from the command line, and many have no graphical interface at all. This section builds the baseline shell literacy required for the rest of the semester. Work through each subsection inside your Kali terminal as you read it.

## 10.1 The Terminal & the Shell

The terminal is a text-based window into the shell — a program (bash, by default in Kali) that reads the commands you type, executes them, and prints their output. Your shell prompt typically shows username@hostname:current-directory$ for a normal user, changing to a # symbol when operating as root, a useful visual reminder of your current privilege level.

## 10.2 The Linux Filesystem Hierarchy

Unlike Windows, Linux has a single unified directory tree starting at the root directory, written as a forward slash: /. Every file, directory, and even device on the system exists somewhere under this single tree — there are no separate drive letters such as C:\ or D:\. Understanding the standard top-level directories helps you know where to look for configuration files, logs, and installed programs.

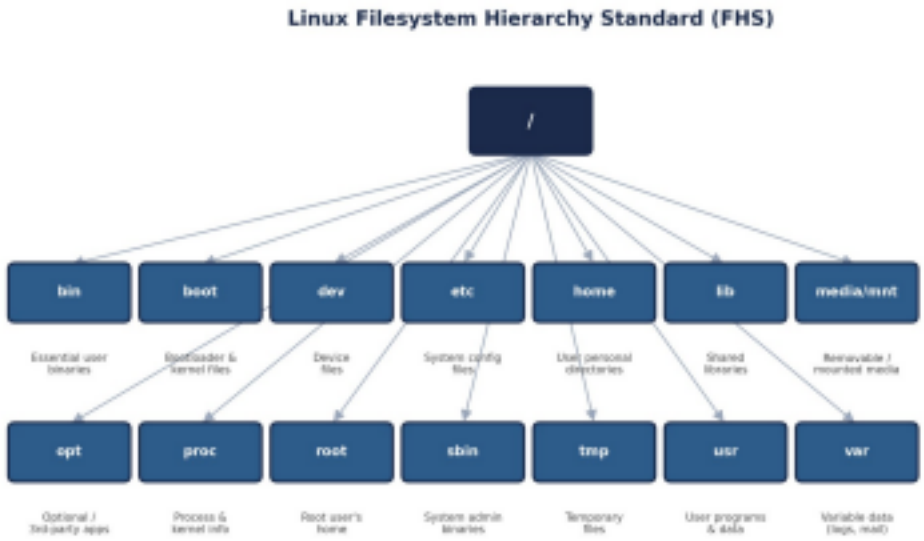



*Figure 10.1 — The standard Linux Filesystem Hierarchy (FHS): key top-level directories and their purpose.*

| | |
|---|---|
| /bin, /usr/bin | Essential executable programs available to all users |
| /etc | System-wide configuration files |
| /home | Personal directories for each regular user |



| | |
|---|---|
| /root | Home directory of the root (superuser) account |
| /var | Variable data: logs, mail spools, caches |
| /tmp | Temporary files, cleared on reboot |
| /dev | Device files representing hardware (disks, USB, etc.) |
| /etc/passwd, /etc/shadow | User account and (hashed) password information |

## 10.3 Navigation Commands

| | |
|---|---|
| pwd | Print Working Directory — shows your current location in the filesystem |
| ls | List the contents of the current directory (ls -la shows hidden files and details) |
| cd <dir> | Change Directory (cd .. goes up one level, cd ~ goes to your home directory) |
| tree | Display a directory's contents as a visual tree (may require: sudo apt install tree) |
| clear | Clear the terminal screen |

## 10.4 File & Directory Operations

| | |
|---|---|
| mkdir <name> | Create a new directory |
| touch <file> | Create a new empty file, or update its timestamp |
| cp <src> <dst> | Copy a file or directory (-r for recursive/directories) |
| mv <src> <dst> | Move or rename a file or directory |
| rm <file> | Remove a file (-r for directories, -f to force — use with caution!) |
| find <path> -name <pattern> | Search for files matching a name pattern |
| locate <name> | Quickly find files using a pre-built index (sudo updatedb to refresh) |



## 10.5 Viewing & Editing Files

| | |
|---|---|
| cat <file> | Print the entire contents of a file to the screen |
| less <file> | View a file one page at a time (q to quit, / to search) |
| head / tail <file> | Show the first / last 10 lines of a file (-n N for a custom count) |
| nano <file> | Beginner-friendly terminal text editor |
| vim <file> | Powerful modal text editor, standard on most Linux/Unix systems |

| | |
|---|---|
| grep <pattern> <file> | Search for lines matching a pattern inside a file |

## 10.6 File Permissions & Ownership

Every file and directory in Linux has an owner, a group, and a set of permissions controlling who may read, write, or execute it. Running ls -l displays these as a ten-character string, e.g., -rwxr xr--, where the first character indicates the type (- for a regular file, d for a directory) and the remaining nine characters are three groups of three, representing permissions for the Owner, Group, and Others, respectively.

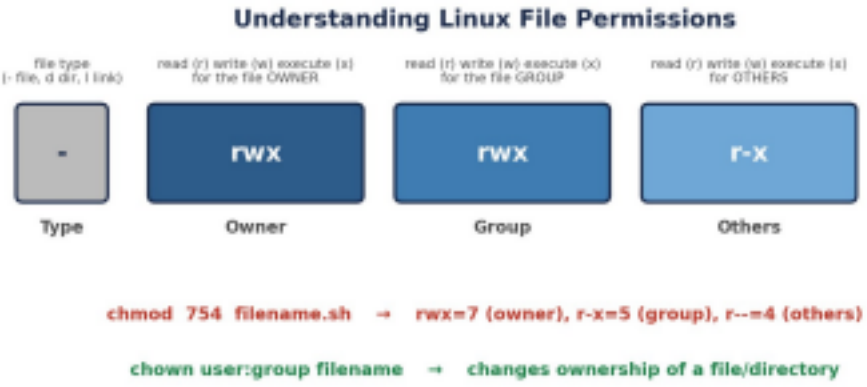



*Figure 10.2 — Anatomy of a Linux permission string and the chmod numeric (octal) notation.*

| | |
|---|---|
| chmod 754 file | Set permissions numerically (owner=rwx, group=r-x, others=r--) |
| chmod u+x file | Add execute permission for the owner (symbolic notation) |
| chown user:group file | Change the owner and group of a file or directory |



| | |
|---|---|
| sudo <command> | Execute a single command with superuser (root) privileges |

## 10.7 Package Management with APT

Kali Linux is Debian-based and uses the Advanced Package Tool (APT) to install, update, and remove software from centrally maintained repositories. Understanding APT is essential, since many later labs will require installing additional tools this way.

| | |
|---|---|

| | |
|---|---|
| sudo apt update | Refresh the local list of available packages and versions |
| sudo apt upgrade | Upgrade all installed packages to their latest available version |
| sudo apt install <pkg> | Install a new package |
| sudo apt remove <pkg> | Remove an installed package (keeps configuration files) |
| sudo apt purge <pkg> | Remove a package and its configuration files |
| apt search <term> | Search for a package by name or description |

### 10.8 User & Process Management

| | |
|---|---|
| whoami | Print the current logged-in username |
| id | Show the current user's UID, GID, and group memberships |
| sudo adduser <name> | Create a new user account |
| ps aux | List all currently running processes |
| top / htop | Interactive real-time view of system processes and resource usage |
| kill <PID> | Terminate a process by its Process ID |

### 10.9 Getting Help

You will not memorize every command and every flag — professionals rely on built-in documentation constantly. Build the habit of checking the manual before searching the web.

```
man <command> # full manual page, e.g. man chmod
```



```
<command> --help # quick summary of options, e.g. ls --help
whatis <command> # one-line description of a command
```



## 11. Hands-On Practice Exercises

Complete the following inside your own Kali Linux VM. These exercises are meant to be

performed, not just read — take a screenshot of your terminal after completing each task if your instructor requires submission.

1. Create a directory called cysec_lab01 inside your home directory, then create three empty files inside it named notes.txt, scan_results.txt, and todo.txt.
2. Use nano or vim to write a one-sentence description of what Cyber Security means to you inside notes.txt, save it, and then display its contents using cat.
3. Change the permissions of scan_results.txt so that only the owner can read and write it, and no one else has any access. Verify the change using ls -l.
4. Use find to locate all .txt files inside your cysec_lab01 directory.
5. Run ps aux and identify at least two processes currently running on your system. Record their Process IDs (PIDs).
6. Update your system using apt and record how many packages were upgraded.
7. Take a VirtualBox/VMware snapshot of your VM and name it "After Lab 01".



## 12. Self-Evaluation Questions

Use the following questions to check your own understanding before moving to the next lab. Try

to answer them without referring back to the manual first.

## 12.1 Multiple Choice Questions

1. Which type of hypervisor runs directly on physical hardware without a host operating system? a) Type-1 (bare-metal) b) Type-2 (hosted) c) Type-3 d) None of the above
2. Which command displays your current working directory in Linux? a) ls b) cd c) pwd d) whoami
3. In the permission string -rwxr-xr--, what access does "Others" have? a) Read only b) Read and write c) Read, write, execute d) No access
4. Which command refreshes APT's local package index without installing anything? a) apt install b) apt upgrade c) apt update d) apt search
5. What is the purpose of a VirtualBox/VMware snapshot? a) Compress the VM's disk b) Save a restorable point-in-time state of the VM c) Update the guest OS d) Back up host files

## 12.2 Short Answer Questions

1. Explain, in your own words, why cyber security practitioners perform experiments inside virtual machines rather than directly on their host operating system.
2. What is the difference between the commands rm and rm -r? Why should rm -rf be used with particular caution?
3. Describe what the command chmod 640 report.txt does, breaking down what each digit means.
4. What is the difference between apt remove and apt purge?
5. Why is it important to verify the SHA256 checksum of a downloaded ISO file before using it?

## 12.3 Practical Self-Check Tasks

- Without looking at the manual, create a directory, create a file inside it, and change that file's permissions so only you can execute it.
- From memory, list the commands you would use to fully update a Kali Linux system.
- Explain to a classmate (or write down) the difference between the root user and a regular user account, and when sudo should be used.

**Self-Assessment Guidance:**

If you could not confidently answer most of the questions above, revisit Section 10 (Linux Command-Line Fundamentals) and repeat the Hands-On Practice Exercises before attempting Lab 02. A solid command of these fundamentals is assumed knowledge for every subsequent lab



in this course.

## 13. Marking Rubric

| | |
|---|---|
| Kali Linux VM successfully installed, updated, and Guest Additions/VMware Tools configured | 30% |
| Hands-On Practice Exercises completed and demonstrated (with screenshots/terminal output) | 35% |
| Self-Evaluation Questions attempted with correct/well-reasoned answers | 20% |
| Snapshot management demonstrated (Clean Install + After Lab 01 snapshots present) | 10% |
| Lab decorum, viva/Q&A responses, and understanding demonstrated to instructor | 5% |



## 14. References

- Official Kali Linux Documentation — https://www.kali.org/docs/
- Kali Linux Downloads — https://www.kali.org/get-kali/
- Oracle VirtualBox User Manual — https://www.virtualbox.org/wiki/Documentation
- VMware Workstation Pro Documentation — https://docs.vmware.com/ ✦ The Linux Documentation Project — https://tldp.org/
- Debian Filesystem Hierarchy Standard — https://www.debian.org/doc/manuals/debian-reference/ch03.en.html